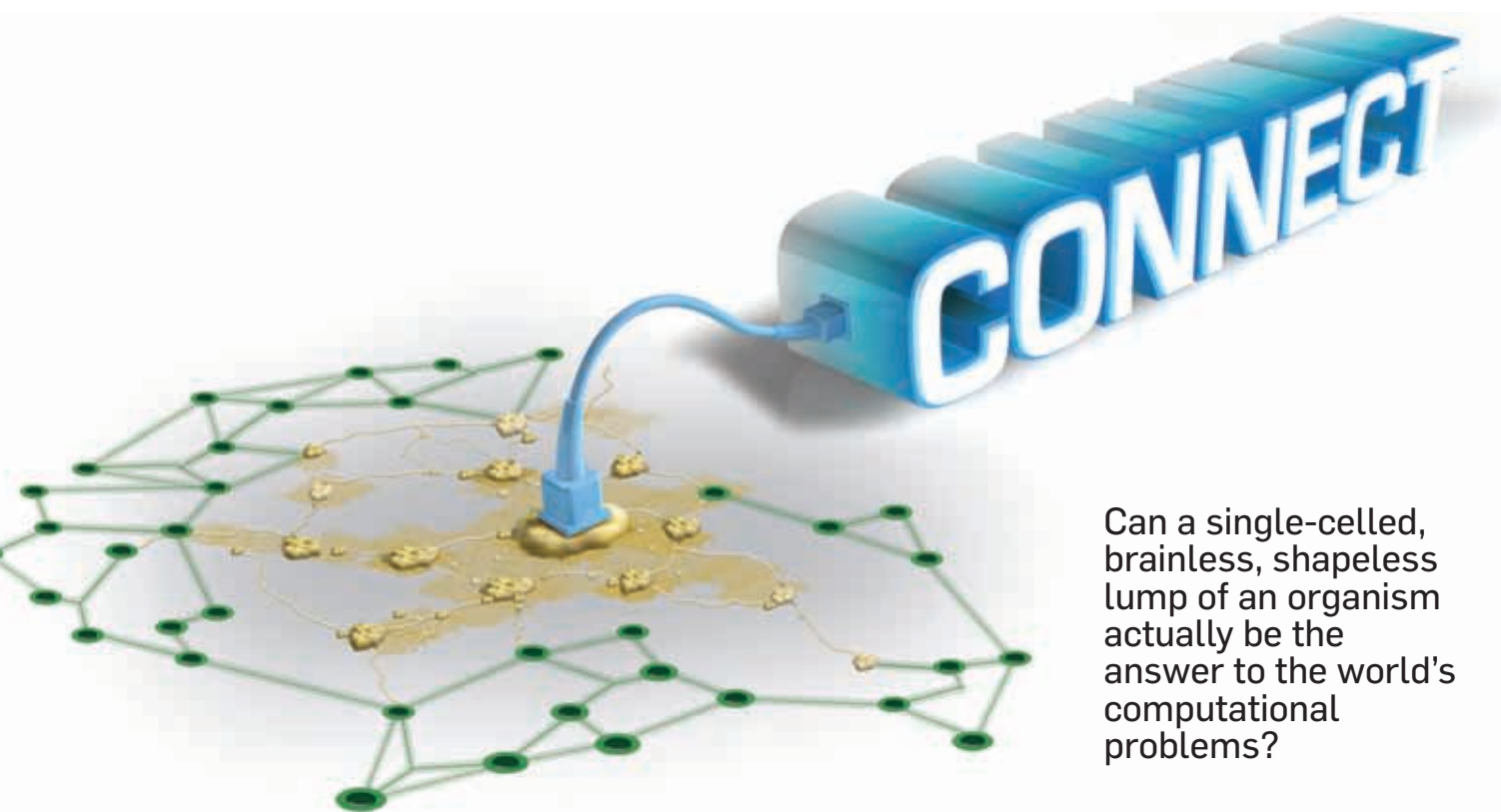


How stuff works

A peek into the internal workings of AMD's Graphics Core Next architecture

**"Intel Inside" mobiles**

Lenovo K800, a smartphone based on Intel Atom Z2460 to be up for sale in Q2 2012
<http://bit.ly/z53CLT>



Can a single-celled, brainless, shapeless lump of an organism actually be the answer to the world's computational problems?

Organic Networking

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Physarum polycephalum or slime mould as it's better known, is a single-celled, amoeba-like organism which prefers to feed on bacteria. What's so special about this, you ask? That will take some explaining.

What does it do?

Slime mould starts life as a small mass of amoeba-like cells, which start multiplying when it senses its favourite food (some say they love sedatives!). When these cells encounter the right mate, they form a 'plasmodia', which is basically one large cell with nuclei floating in the cytoplasm (the life 'blood' of these organisms). The slime mould starts pulsing, similar to your heart pulsating as it pumps blood. Every time the mould expands, it feels or senses around for more food. When it

encounters food, it will start to pulse more in that direction and less in the others. This process repeats itself till the mould has enveloped all the food in its vicinity. By the end of this process, the mould can grow up to a metre or more in diameter. Think about it, a single-celled organism that is larger than a pizza!

Once this slime mould is satisfied that it has got all the food in the vicinity, all the nuclei start receding and settle down in the centre of the mould. The rest of the mould also recedes into the centre, leaving behind very thin protoplasmic strands (similar to veins), connecting all the food sources via a complex network and ensuring the most

efficient way to transport the food to the central core. If a strand is carrying a larger volume of food, it will expand, if it isn't it will contract, eventually to disappear. The cytoplasm in these 'veins' is responsible



Slime mould in nature

IMAGING: ATUL DESHMUKH